



Jahangirnagar University
Department of Computer Science and Engineering
4th Year 1st Semester B.Sc. (Hons.) Final Examination -2023

Course Title: **Digital Image Processing**

Course No: **CSE-405**

Time: **3.00 hours**

Full Marks: **60**

[Answer each of the following questions. Each question carries equal marks. Figures in the right margin indicate marks.]

1. Answer *All of them*

- a) Define brightness and contrast of an image. 2
- b) Define image restoration and noise (in image). 2
- c) State some of the applications of digital image processing in different fields. 2
- d) Mention the general application areas of CMY, YIQ and HSI color models. 2
- e) List approaches for dealing with the problem of applying a filter at the border of an image. 2
- f) Define (*any two*): 2
 - i. Lookup Table (LUT)
 - ii. FFT
 - iii. 2-D Moment (of an image)

2. Answer *Any Three* out of *Four*.

- a) Compare *unsharp masking* and *high boost filtering*. 4
- b) What are the sources of noise in digital images? Also discuss a noise model accommodating different types of noises. 4
- c) Point out the basic principle of histogram equalization, histogram modeling and histogram specification for image enhancement. 4
- d) Explain the relation between RGB, HSI and CMYK. 4

3. Answer *Any Three* out of *Four*.

- a) Describe the Otsu method of adaptive thresholding. 4
- b) i. Explain the Laplacian filtering technique. 4
- ii. Define Derivative filter. 4
- c) Illustrate the homomorphic transformation for image enhancement. 4
- d) Write a short note on image segmentation techniques. 4

4. Answer *Any Three* out of *Four*.

- a) Classify image compression methods and explain *run length encoding (RLE)* with an example. 4
- b) Prove that first order moment invariants, $\mu_{10} = \mu_{01} = 0$. Why we do not use higher and lower-order moments in image processing? 4
- c) Write the thinning algorithm. 4
- d) Write the spatial mask for the following 4
 - i. Prewitt Mask
 - ii. Laplacian Mask

5.

Answer *Any Two* out of *Three*.

- a) i. Consider a 5-bit grayscale image of resolution 6×7 is shown below in Fig.1. Draw its histogram and CDF. Comment on the quality of this image if it is represented using 8-bit (on the basis of a linear scale) then express your opinion regarding the image. 6

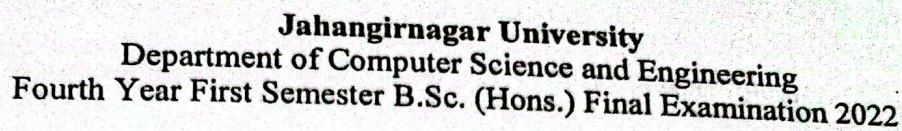
0	5	4	10	10	0	0
6	5	5	4	4	4	0
0	20	2	2	2	4	4
30	0	2	2	2	4	4
0	0	2	2	2	4	4
0	10	3	3	20	31	0

Fig.1

- ii. Convert the image shown in Fig.1 into its binary equivalent using a threshold value of 15. 6
- b) Summarize the steps to evaluate principal component analysis (PCA) of a given dataset. 6
- c) Perform erosion operation for the image shown in the following Fig.2. Use a 3×3 cross-size structuring element. 6

0	0	0	1	1
0	1	1	0	1
0	1	0	0	1
1	1	1	1	1
1	1	0	1	1

Fig.2



Course No: CSE-405
Time: 3 Hrs.

1. Answer all questions:

- | | |
|--|---|
| a) Define (<i>any two</i>): i) Image processing ii) Noise (in image) | 2 |
| iii) Zero crossings iv) Wavelet | |
| b) How can we differentiate between illumination and reflectance? | 2 |
| c) Sketch the match band effect. | 2 |
| d) State some of the applications of digital image processing in different fields. | 2 |
| e) Define (<i>any two</i>): | 2 |
| i) Color map ii) FFT iii) Bit-plane iv) Discrete Laplacian | |
| f) What is image histogram? | 2 |

2. Answer Any Three out of Four questions:

- | | |
|--|---|
| a) Explain SURF/SIFT image feature descriptor. | 4 |
| b) Describe different energy sources used in creating images. | 4 |
| c) Discuss the reasons for the importance of Gaussian filters. | 4 |
| d) Explain the relation between RGB, HSV, CMYK and YCbCr. | 4 |

3. Answer Any Three out of Four questions:

- Answer Any Three out of Four questions.
- a) Write a complete Python/MATLAB program to perform (any ONE):
- i) Histogram of a color (RGB) image
 - ii) Average filtering of a gray scale image
 - iii) Cleaning salt and pepper noise from a noisy grayscale image using median filtering.
- b)
- i) Differentiate between brightness and contrast.
 - ii) Explain the convolution property in 2D fourier transform.
- c)
- i) Differentiate linear filter and non-linear filter.
 - ii) State the importance of hue and saturation in the formation of color image.
- d) Construct a model of image degradation.

4. Answer **Any Three** out of **Four** questions:

- Answer **Any Three** out of **Four** questions:
- a) Classify image compression methods and explain Huffman coding with an example.

- b) Explain *ROI (region of interest) filtering* in detail. 4
 - c) Distinguish among 2-D moment, central moment, and normalized central moment of order $(p+q)$. 4
 - d) Discuss the trade-offs between various filtering techniques in terms of i) noise reduction 4
ii) detail preservation
5. Answer **Any Two** out of **Three** questions:
- a) Compare between single and double thresholding method for image segmentation and summarize the applications of thresholding. 6
 - b) Summarize the steps to evaluate principal component analysis (PCA) of a given dataset. 6
 - c) Write a complete Python/MATLAB program that reads a color (RGB) image and then performs edge detection on this image. 6